

Part-of-Speech Tagging using a Smoothed Bigram HMM

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MFS Data Science and Artificial Intelligence

August 27, 2026

Abstract

Part-of-speech (POS) tagging is the task of assigning a grammatical category to every word in a sentence. We have to build a model based on bigram assumption that handles "zero-events" probabilities and out-of-vocabulary events.

The final model achieves an accuracy of **96.4080%** on the **dev.sents** dataset, corresponding to 45,922 correctly tagged tokens out of 47,633. The model achieves 96.70% accuracy on known words and 85.44% accuracy on OOV words. Model training and development-set tagging take **4.46 s** and **6.88 s**, respectively.

1 Introduction

In this assignment, we implement a Hidden Markov Model (HMM) based POS tagger. The model uses a **bigram assumption**: the POS tag of the current word depends on the POS tag assigned to the preceding word. The model therefore combines two sources of information:

1. the likelihood of transitioning from one POS tag to another, and
2. the likelihood of a word being generated by a particular POS tag.

The main difficulty beyond the standard HMM formulation is handling **out-of-vocabulary (OOV)** words. Since no direct word-level emission statistics are available for unseen words, the implemented system uses orthographic shape and suffix statistics learned from infrequent training words.

The goal of the experiments was to determine a configuration that improves OOV performance while preserving the strong performance of the underlying HMM on known words.

2 Dataset

The dataset consists of sentences in a word/tag format, where each token is annotated with a POS tag. The training set is used to estimate the HMM parameters and the development set is used for evaluation and error analysis.

The development set contains:

- **47,633** total tokens,
- **46,417** known tokens,
- **1,216** OOV tokens.

Thus, OOV tokens account for approximately 2.6% of the development set.

Table 1: Development-set statistics.

Statistic	Count
Total tokens	47,633
Known tokens	46,417
OOV tokens	1,216

3 Methodology

We implement a first-order Hidden Markov Model (HMM) for POS tagging. Each sentence is represented as a sequence of observed words $w_1, \dots, w_n$ and hidden POS tags $t_1, \dots, t_n$. The model estimates the most likely tag sequence using transition and emission probabilities.

The transition model captures dependencies between consecutive tags:

$$P(t_i | t_{i-1}),$$

while the emission model estimates the likelihood of observing a word given its tag:

$$P(w_i | t_i).$$

For a sentence, the best tag sequence is obtained using the Viterbi algorithm. Probabilities are computed in log space to avoid numerical underflow.

Words not observed in the training vocabulary are treated as OOV words. For these words, the model supplements the standard emission estimate with word-shape and suffix-based information. The OOV model was developed through a second-pass rescoring procedure, with its experiments described in Section 5.

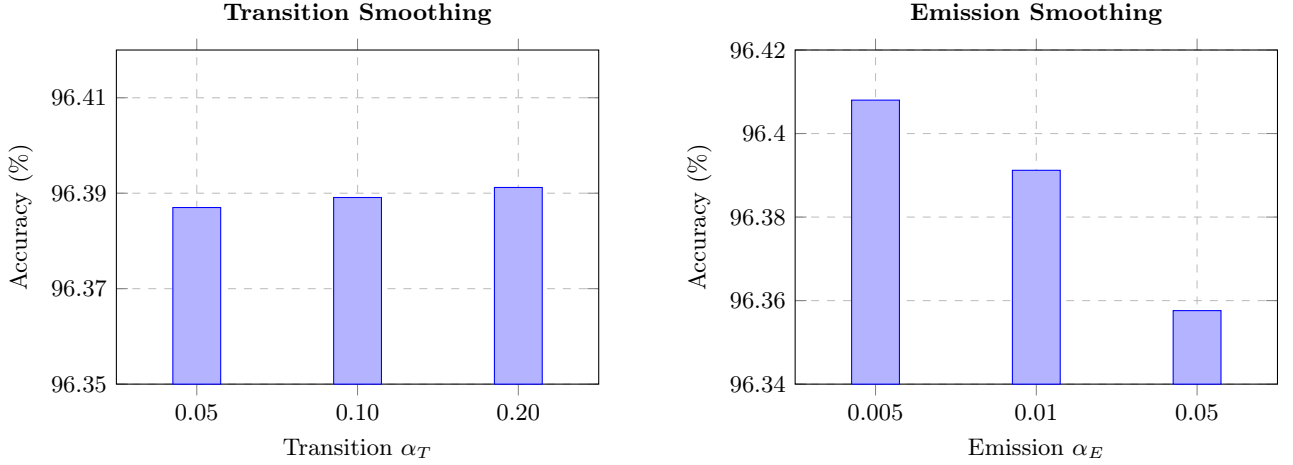


Figure 1: Effect of smoothing constants on development-set accuracy. Transition smoothing was evaluated at $\alpha_T \in \{0.05, 0.10, 0.20\}$, while emission smoothing was evaluated at $\alpha_E \in \{0.005, 0.01, 0.05\}$.

4 Smoothing

Additive smoothing was applied to both transition and emission probabilities to avoid zero-probability events:

$$P(t_i | t_{i-1}) = \frac{C(t_{i-1}, t_i) + \alpha_T}{C(t_{i-1}) + \alpha_T(|T| + 1)},$$

$$P(w_i | t_i) = \frac{C(t_i, w_i) + \alpha_E}{C(t_i) + \alpha_E(|V| + 1)}.$$

The smoothing constants were tuned on the development set. Among the tested values, $\alpha_T = 0.20$ and $\alpha_E = 0.005$ produced the best overall accuracy of **96.4080%**. The complete parameter comparison is given in Table 2.

5 OOV Handling

OOV words cannot use their ordinary word-specific emission counts because the corresponding word is absent from the training vocabulary. The model therefore estimates the POS distribution of an unseen word from surface-level information.

5.1 Word Shape

The implementation identifies several word-shape classes:

- lowercase words,
- capitalized words,
- all-capital words,
- hyphenated words,
- words containing digits.

These features provide useful information about the likely POS tag. For example, capitalization can provide evidence for proper nouns, while hyphenation and digits can provide additional morphological or orthographic clues.

5.2 Rare-Word Simulation

To make the training data more representative of OOV words, shape and suffix statistics are learned from relatively infrequent words.

A training word contributes to this component if its corpus frequency satisfies

$$f(w) \leq R,$$

where the final value was

$$R = 4.$$

This allows the system to obtain morphological information from words that are less frequent in the training corpus while avoiding dependence only on fully observed common words.

5.3 Suffix Modeling

For each OOV word, the model estimates tag distributions from suffixes of up to five characters, retaining only suffixes observed at least twice in the training data. Suffix evidence is interpolated with the broader word-shape distribution using $w = N/(N + \lambda)$, where N is the suffix count and $\lambda = 20$. This conservative weighting limits the influence of sparse suffix statistics.

5.4 Second-Pass OOV Rescoring with Future Context

We also experimented with using future-tag context $P(t_{i+1})$ during second-pass OOV rescoring to exploit both left and right context. However, this reduced development-set accuracy, so the future-tag component was removed from the final model. This may be due to sparse and noisy estimates arising from the larger contextual space and limited training observations.

Table 2: Summary of parameter experiments.

Parameter	Initial	Selected	Observation
Rare-word limit	2	4	Improved OOV accuracy.
Minimum suffix observations	2	2	Increasing to 3 reduced accuracy.
Maximum suffix length	6	5	5 gave better OOV performance than 4 and 6.
Suffix backoff strength	5	20	Increasing the backoff strength improved OOV accuracy.
Transition smoothing α_T	0.1	0.2	0.2 gave the highest overall accuracy among tested values.
Emission smoothing α_E	0.01	0.005	0.005 gave the highest overall accuracy among tested values.

6 Experimental Evaluation

We performed controlled experiments to tune the OOV model and the HMM smoothing parameters. Each parameter was varied while keeping the remaining parameters fixed at their current best values.

The final configuration obtained from these experiments is:

$$\alpha_T = 0.2, \quad \alpha_E = 0.005, \quad R = 4, \\ L = 5, \quad M = 2, \quad \lambda = 20$$

where R denotes the rare-word limit, L the maximum suffix length, M the minimum suffix observations, and λ the suffix backoff strength.

The initial model achieved 96.3765% overall accuracy and 84.79% OOV accuracy. With the selected configuration, performance increased to 96.4080% overall accuracy and 85.44% OOV accuracy.

7 Final Results

The final model achieves:

Table 3: Final development-set performance.

Metric	Result
Correct tokens	45,922 / 47,633
Overall accuracy	96.4080%
Known-word accuracy	96.70%
OOV accuracy	85.44%

Compared with the initial baseline, overall accuracy increased from 96.3765% to 96.4080%. OOV accuracy improved more substantially, from 84.79% to 85.44%.

8 Error Analysis

The final model was evaluated separately on words observed in the training vocabulary and out-of-vocabulary (OOV) words. This distinction provides

a useful view of where the model performs well and where its main limitations remain.

8.1 Known and OOV Performance

The final model achieves an accuracy of **96.70%** on known words, compared with **85.44%** on OOV words. This gap indicates that unseen words remain substantially more difficult to tag than words for which word-level emission statistics are available.

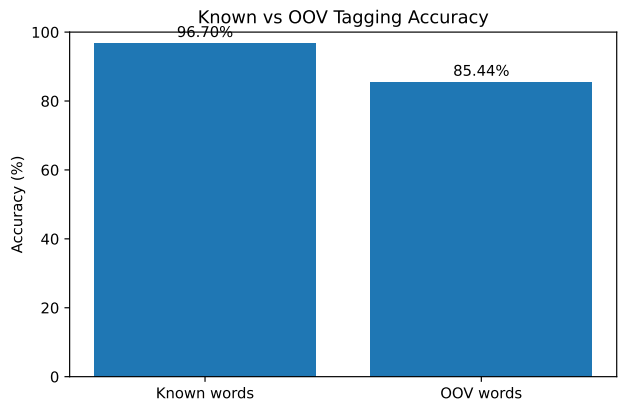


Figure 3: Tagging accuracy on known and OOV words.

The OOV accuracy nevertheless improved from 84.79% for the initial model to 85.44% in the final configuration. This improvement is consistent with the benefit obtained from the suffix- and shape-based OOV model described earlier.

8.2 Confusion Matrix Analysis

The largest observed confusion is **NN** $\rightarrow$ **JJ**, with 197 errors, followed by **VDN** $\rightarrow$ **JJ** with 115 errors and **NN** $\rightarrow$ **NNP** with 95 errors. The **VBD** $\rightarrow$ **VDN** confusion is also prominent, with 82 errors.

A further breakdown shows that many of these errors occur on words that are already present in the training vocabulary. For example, all 82 **VBD** $\rightarrow$ **VDN** errors are known-word errors, while 113 of the 115 **VDN** $\rightarrow$ **JJ** errors are also known-word errors. Similarly, the

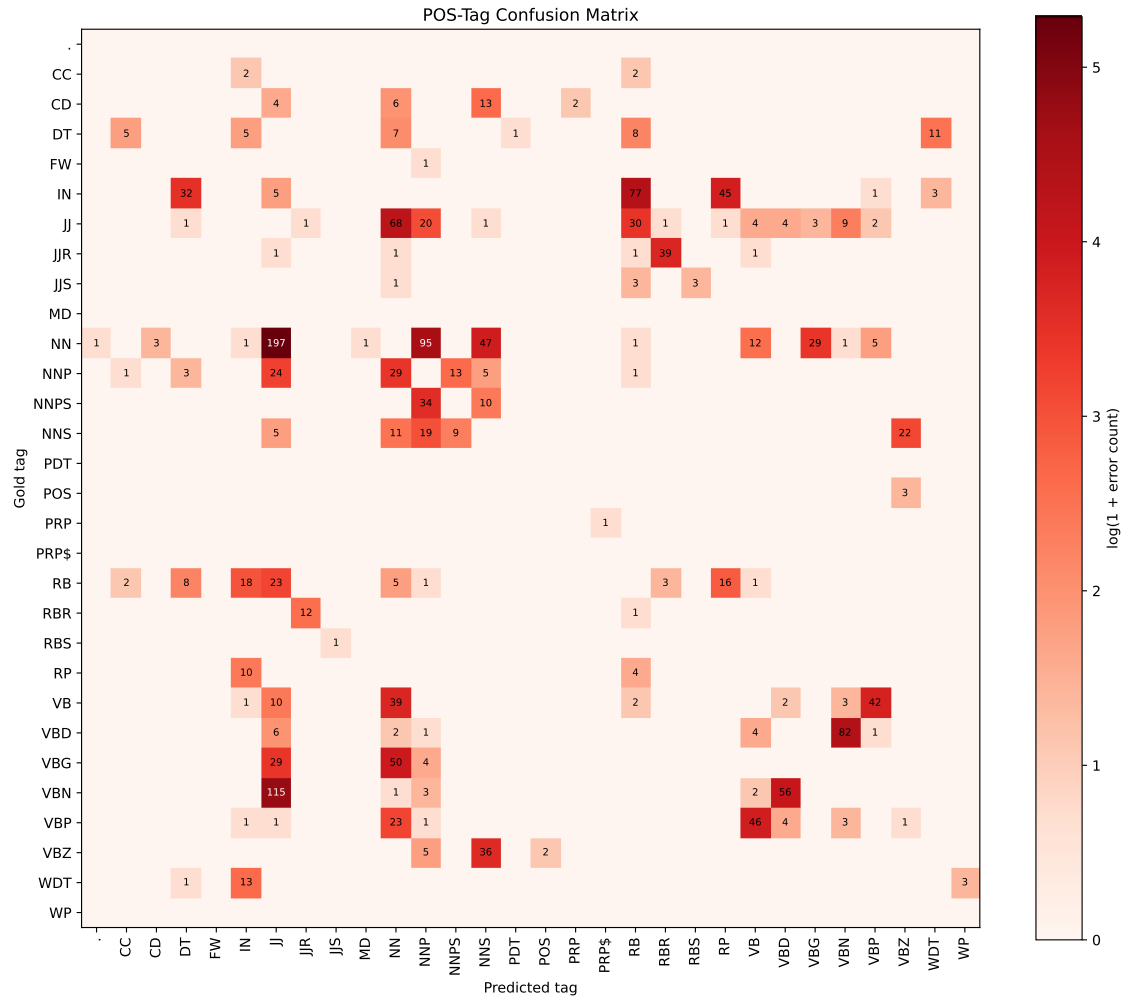


Figure 2: POS-tag confusion matrix for the final model. Rows represent gold tags and columns represent predicted tags. Correct classifications are omitted; darker cells indicate larger error counts.

197 **NN** $\rightarrow$ **JJ** errors consist of 162 known and 35 OOV errors.

This suggests that although OOV handling remains an important source of error, the dominant remaining limitations of the model are not exclusively caused by unseen words. In particular, ambiguities between syntactically similar tags require contextual information that is not always captured by a first-order bigram HMM.